Module 10: Epigenetics — Study Questions

1. What does it mean for a gene to be "expressed"?
2. Every cell in your body has the same DNA. Why, then, do muscle cells and nerve cells look and act differently?
3. What is a transcription factor, and what role does it play in gene regulation?
4. In your own words, define epigenetics. How is an epigenetic change different from a mutation?
5. What is DNA methylation? Does it generally turn a gene on or off?
6. What are histones? What happens to gene activity when histones are acetylated (have acetyl groups added)?
7. Using the memory trick from class: **M**ethylation = **M**uting, **A**cetylation = **A**ctivation. Explain in a sentence why each association makes sense.
8. What is X-inactivation? What is a Barr body?
9. How do calico cats demonstrate X-inactivation? Why don't male calico cats normally exist?
10. Give two examples of how the environment (diet, stress, exercise, or toxins) can affect which genes are turned on or off.
11. Why might identical twins develop different traits or health conditions as they get older, even though they share the same DNA?
12. Can epigenetic changes be reversed? How does this differ from a permanent DNA mutation?
13. **Putting it all together:** A cell in your liver and a cell in your eye have identical DNA. Using what you learned about gene regulation and epigenetics, explain how these two cells can have completely different structures and functions.